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23K-0008

Agentic AI Assignment Date:

Beta

Ques #03

(a) $p(i) = \exp(z_i) / \sum_j \exp(z_j)$

Teacher $z_i = [5, 3, 2, 0]$.

$\exp(5) = 148.413159$, $e^3 = 20.085537$

$e^2 = 7.389056$, $e^0 = 1$.

$\Sigma = 148.413159 + 20.085537 + 7.389056 + 1$

$\Sigma = 176.887752$.

$P(1) = 148.413159 / 176.887752 = 0.839025$.

$P(2) = 20.085537 / 176.887752 = 0.113550$.

$P(3) = 7.389056 / 176.887752 = 0.041773$.

$p(4) = 1 / 176.887752 = 0.005653$

$P_t = [0.839025, 0.113550, 0.041773, 0.005653]$.

$z_s = [3, 2.5, 1, 0.5]$.

$e^3 = 20.085537$, $e^{2.5} = 12.182494$, $e^1 = 2.718282$

$e^{0.5} = 1.648721$.

$\Sigma = 20.085537 + 12.182494 + 2.718282 + 1.648721$.

$\Sigma = 36.635034$.

$q(1) = 0.548260$, $q(2) = 0.332537$

$q(3) = 0.074199$, $q(4) = 0.045004$.

$q_t = [0.548260, 0.332537, 0.074199, 0.045004]$.

(b) Softened probabilities :-

$p_T(i) = \exp(z_i / T) / \sum_j \exp(z_j / T)$.

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Teacher $z_t/2 = [2.5, 1.5, 1, 0]$.

exp = [12.182494, 4.481689, 2.718282, 1].

$S_1 = 20.382465$.

P_T(1) = 0.597695, P_T(2) = 0.219880

P_T(3) = 0.133364, P_T(4) = 0.049062.

P_T = [0.597695, 0.219880, 0.133364, 0.049062].

Students $z_s/2 = [1.5, 1.25, 0.5, 0.25]$.

exp = [4.481689, 3.490343, 1.648721, 1.284025].

$S_1 = 10.904779$.

q_T(1) = 0.410984, q_T(2) = 0.320075.

q_T(3) = 0.151193, q_T(4) = 0.117749.

q_T = [0.410984, 0.320075, 0.151193, 0.117749].

(c) KL divergence.

formula:- $KL(P_T \| q_T) = \Sigma p_T(i) \cdot \ln(p_T(i)/q_T(i))$

Word	P_T/q_T	ln(ratio)	p_T * ln(ratio)
hospital	1.454302	0.374526	0.223852
clinic	0.686964	-0.375474	-0.082559
ward	0.882079	-0.125474	-0.016734
school	0.416665	-0.875474	-0.042952

KL = 0.223852 - 0.082559 - 0.016734 - 0.042952

[KL = 0.081607]

(d) Hard label cross-entropy:-

$CE(y, q_i) = -\Sigma y(i) \cdot \ln(q_i)$.

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$CE = -[1 \cdot \ln(0.548260) + 0 \cdot \ln(0.332537) + 0 + 0]$.

$\ln(0.548260) = -0.601005$.

$CE = -(-0.601005)$:

$\boxed{CE = 0.601005}$

(e) Total distillation loss:-

$L = \alpha T^2 \cdot KL + (1-\alpha) CE$.

$\alpha \cdot T^2 = 0.5 \times 2^2 = 2$.

$\alpha T^2 \cdot KL = 2 \times 0.081607 = 0.163215$

$(1-\alpha) \cdot CE = 0.5 \times 0.601005 = 0.300502$.

$\boxed{L = 0.463717}$

(f) Gradient w.r.t student logits.

$\frac{\partial L}{\partial z_s} = \alpha \cdot T \cdot (q_T - p_T) + (1-\alpha)(q_1 - y)$.

$\alpha \cdot T = 0.5 \times 2 = 1$, $(1-\alpha) = 0.5$.

Soft term $= 1 \times (q_T - p_T)$.

$i = 1$; $0.410984 - 0.597695 = -0.186711$.

$i = 2$; $0.320075 - 0.219880 = 0.100195$.

$i = 3$; $0.131193 - 0.133364 = 0.017829$.

$i = 4$; $0.117749 - 0.049062 = 0.068687$.

Hard term $= 0.5 (q_1 - y)$:

$i = 1$; $0.5(0.548260 - 1) = -0.225870$.

$i = 2$; $0.5(0.332537) = 0.166268$.

$i = 3$; $0.5(0.074199) = 0.037099$.

$i = 4$; $0.5(0.045004) = 0.022502$.

Sum- $\partial L / \partial z_{s1} = -0.186711 - 0.225870 = -0.412581$

$\partial L / \partial z_{s2} = 0.100195 + 0.166268 = 0.266463$.

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$\partial L/\partial z_{53} = 0.017829 + 0.037099 = 0.054928.$

$\partial L/\partial z_{54} = 0.068687 + 0.022502 = 0.091189.$

$\partial L/\partial z_5 = [-0.412581, 0.266463, 0.054928, 0.091189].$

Ques #04.

(1) Accuracy = $\frac{230}{300} = 0.7667$. Stage 2: $\frac{254}{300} = 0.8467$.

$d = 1 + z^2/n = 1.012805.$

Centre = $(\hat{p} + z^2/2n)/d$. Stage 1: 0.76330

Stage 2: 0.84228.

Half width = $z\sqrt{(\hat{p}(1-\hat{p})/n + z^2/4n^2)}/d$

Stage 1: 0.04768 Stage 2: 0.04075.

CI Stage 1: [0.7156, 0.840], Stage 2: [0.8015, 0.8830].

(2) Per-class ($P = TP/(TP+FP)$, $R = TP/(TP+FN)$, $F1 = 2PR/(P+R)$).

Stage 1	TP	FP	FN	P	R	F1
Pos	100	28	20	0.781	0.8333	0.8065
Neg	80	15	20	0.8421	0.8	0.8205
Neu	50	27	30	0.6494	0.6250	0.6369.

Stage 2	TP	FP	FN	P	R	F1
Pos	106	15	14	0.8760	0.8833	0.8797
Neg	84	11	16	0.8842	0.8400	0.8615
Neu	64	20	16	0.7619	0.8	0.7805.

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③ Averages:-

	Stage 1	Stage 2
Macro	0.9546	0.8406
Weighted	0.7659.	0.8472
Micro	P=R=F1=0.7667	0.8467.

④ Interpretation:-

- Best class: Negative (F1=0.82). Weakest: Neutral (F1=0.64).
- Neutral is confused with Positive (20) and Negative (10)! Pos & Neg are wrongly sent to Neutral 15 & 12 times.
- Neutral has no clear priority and prompt did not define it.
- Fix: define Neutral and add Neutral few shot examples.

⑤ Comparison:-

- Everything improved: Accuracy: +0.080, Macro F1: +0.086, Weighted F1: +0.081.
- The biggest gain is Neutral recall (+0.175) and Neutral F1 (+0.144). Negative improved least (+0.04).
- The CIs overlap and both stages use the same 300 reviews, so no significance claim.

⑥ Stability:-

Mean ± 4.227/5 = 0.0454.

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- Deviations = +0.0016, −0.0054, +0.0076, −0.0084, +0.0046.
- Squares = 0.00002256, 0.00002916, 0.00005776, 0.00007056, 0.00002116.
  Sum = 0.0001812.
- Variance = 0.0001812/4 = 0.0000453.
- S = √0.0000453 = 0.0067.
- Result = 0.8454 ± 0.0067, so the prompt is stable.

(7) BLEU/ROUGE/METEOR:-

- BLEU-4 : p1 = 8/9, p2 = 5/8, p3 = 3/7, p4 = 2/6.
- BP = exp(1 − 10/9) = 0.8948.
- BLEU-4 = 0.8948 × 0.5307 = 0.4750.
  - No smoothing was needed, bcz no precision is 0.
- ROUGE-1: P = 8/9; R = 8/10; F1 = 0.8421.
- ROUGE-2: P = 5/8, R = 5/9; F1 = 0.5882.
- ROUGE-L LCS: "the product quality is good" = 5
  P = 5/9, R = 5/10, FL = 0.5263.
- METEOR (NLTK, α = 0.9, β = 3, γ = 0.5, exact stem, WordNet: off):
  m = 8, P = 8/9, R = 0.8.
  Fmean = PR / (0.9P + 0.1R) = 0.8081.
  chunks = 3, penalty = 0.5(3/8)² = 0.0237.
  METEOR = 0.8081 × (1 − 0.0237) = 0.7868.

BLEU & ROUGE compare surface strings, so "late" & "slow" gets no credit and the swapped clause order lowers the scores.

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(8) Perplexity:-

$\sum \ln(p) = -7.5929$.

Average negative log prob: $7.5929/9 = 0.8437$.

PPL $\Rightarrow e^{0.8437} = 2.325$.

The model was as unsure as choosing among ~2.3 equally likely tokens. This is not classification accuracy.